HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed **Python** tools for automated data acquisition and analysis, reducing manual data collection time by **40%**.
- Designed and implemented **embedded C firmware** for microcontrollers, focusing on efficient memory management and real-time task execution.
- Integrated **CAN and TCP/IP** communication interfaces for inter-device data exchange and network connectivity in embedded systems.
- Debugged and brought up new **evaluation boards and embedded platforms** in a lab environment, resolving hardware integration issues.
- Implemented and validated **battery algorithms** including SoC and SoH estimation, contributing to improved battery performance monitoring.
- Supported customer evaluations by providing technical guidance and troubleshooting, ensuring successful integration of battery solutions.
- Managed **Linux** server configurations and shell scripting for deployment automation and log analysis in development environments.

SKILLS

Languages – Python, C, SQL, Java, Shell Scripting.

Embedded Systems – Microcontrollers, ARM Cortex-M, BMS, ESS, Cell Monitoring, SoC, SoH, Kalman Filtering, Signal Processing.

Communication Protocols – I2C, SPI, UART/USART, CAN, TCP/IP.

Tools & Platforms – Git, Linux, Docker, Jenkins, Jira, Confluence, AWS (EC2, S3, Lambda, CloudWatch), Oscilloscope, Logic Analyser, Multimeter.

Methodologies – Agile, Scrum, Test Automation, Data Acquisition, Data Analysis.

Hardware – Schematics, ADCs, Analog Front Ends, Evaluation Boards, Embedded Platforms.

PROJECTS

Embedded Battery Monitoring System Development

- Developed **Python** scripts to automate data logging from embedded devices via **UART**, capturing over **50,000** data points per test run for algorithm validation.
- Wrote **embedded C** firmware for a microcontroller to manage battery cell balancing and collect state-of-charge (SoC) data, supporting a **12-cell** battery pack.
- Integrated **I2C** communication for sensor readings and **CAN** for reporting system status to a central gateway, ensuring reliable data flow.

Automated Test Framework for Embedded Peripherals

- Engineered a **Python**-based test framework to interact with embedded hardware through **SPI** and **TCP/IP**, automating the verification of peripheral functionalities.
- Designed test cases to validate communication interfaces and data integrity, reducing manual testing effort by **30%** and improving test coverage.
- Implemented logging and reporting mechanisms within the framework to track test execution status and identify regressions.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)